

MediKiosk SIH26047 - Current Code vs Longitudinal Requirements

Evidence-backed handoff audit and migration direction

Audience: project owner and implementation team

Audit date: 24 August 2026

Scope: /Users/jagpreet/Documents/Vibe_Coding/SIH_test compared with the complete handoff /Users/jagpreet/Downloads/SIH26047_CODEX_HANDOFF_FULL_CONTEXT_AND_AUDIT.md.

Executive answer

The code is a strong, test-backed **single-encounter intake and physician-draft system**, not yet the handoff's **longitudinal clinical-memory product**.

Using the handoff's exact 40-capability checklist and equal weighting:

- DONE: 12/40 (30%)
- PARTIAL: 22/40 (55%)
- MISSING: 4/40 (10%)
- UNSAFE: 2/40 (5%)
- Weighted implementation maturity (DONE=1, PARTIAL=0.5, MISSING/UNSAFE=0): **57.5%**
- Weighted gap to target: **42.5%**

The headline hides an important asymmetry. The current-encounter intake foundation is roughly three-quarters mature, while the eight-capability longitudinal spine is approximately **25% mature** (four partial, four missing). The main work is therefore not "improve the chatbot"; it is to introduce durable Patient -> Encounter -> source-linked event history and rebuild the doctor view around it.

1. What the code actually does today

The React kiosk runs language selection, mock ABHA login, granular consent, an adaptive YAML-driven interview, optional browser speech, document upload/OCR, patient review, and submission. The physician surface shows a priority queue, a deterministic traceable draft summary, per-session document timeline, contradictions, red flags, fact editing, FHIR preview, and whole-encounter commit.

The interview is not a fixed 28-question form. `DialogueMachine.next_question()` evaluates deterministic `ask_if` rules; LLM/rule extraction does not choose the next question. Every accepted answer is intended to pass through `record_fact()` with a source span. Summary generation is deterministic by default and rejects untraceable claims.

On physician commit the system generates a FHIR bundle, optionally posts it to the local HIS stub, stores the JSON bundle, and purges the session's structured facts and document metadata. This lifecycle is correct for temporary capture but does not promote confirmed facts into a durable longitudinal record.

2. Exact current architecture

- Frontend: React 18, TypeScript, Vite, React Router; routes /, /intake, /physician, /demo.
- Backend: FastAPI, Pydantic v2, async SQLAlchemy.
- Persistence: SQLite by default; Postgres/Redis compose options. Redis or in-process cache holds session state.
- Clinical state: IntakeSession, SessionFact, SessionDocument, RedFlagProposal are session-scoped. ConsentRecord, AuditEvent, terminology, and SubmittedBundle are durable.
- AI: offline rule extractor by default only when no Groq key is selected; optional Groq extraction/prose smoothing; deterministic workflow/rules remain authoritative.
- OCR: text-layer and Tesseract implementations behind a protocol.
- Speech: browser Web Speech in the shipped UI; optional server adapters exist.
- Interoperability: FHIR bundle builder and a same-process demo HIS receiver.

3. Current user flow

Landing -> language -> mock ABHA/OTP -> consent -> adaptive questions -> optional documents -> patient review/correction -> physician queue -> draft summary/source/red flags/conflicts/current-session timeline -> edit -> FHIR preview -> confirm/commit -> session purge.

There is no patient lookup that reconstructs prior encounters for a new intake, and no doctor view that retrieves a patient's prior structured history.

4. Current data flow

Tap/text: frontend answer -> POST /dialogue/answer -> record_answer() -> record_fact() -> session ledger + DB SessionFact -> deterministic red flags -> next state-machine question.

Voice: browser transcript/confidence -> POST /dialogue/answer/voice -> extraction gates -> record_fact() -> red flags -> next question. No audio bytes are posted.

Document: upload -> OCR -> regex entity extraction -> confident facts or verification lane -> session document JSON -> per-session timeline.

Summary: session ledger -> projection -> deterministic assembly -> traceability gate -> physician view -> FHIR builder -> stored SubmittedBundle JSON -> purge session facts/documents/cache.

5. Current data model and longitudinal feasibility

There is no durable Patient, Encounter, ClinicalFact, SourceEvidence, MedicationEvent, ObservationEvent, TimelineEvent, or similarity model. abha_ref is repeated on sessions and bundles, but new

sessions do not load old bundles by patient. The only durable clinical artifact is an opaque FHIR bundle JSON indexed by `session_ref` and `abha_ref`.

This means longitudinal history is theoretically recoverable by future bundle parsing, but **not operationally available today**. There are no patient-history endpoints, same-patient encounter queries, cross-visit medication reconciliation, recurrence logic, trend queries, or similarity retrieval.

6. Real vs mocked or unverified

Real: adaptive state machine, provenance checks, offline extraction, deterministic red flags, contradiction detection, OCR engines/entity parser, traceable summary, consent/purge mechanics, FHIR generation/preview, audit hash chain, synthetic evaluation.

Mock/stub: ABHA identity and OTP, staff login, HIS receiver, patient fixtures, small terminology subsets.

Optional/unverified: live Groq LLM/Whisper, BHASHINI live endpoint, Vosk model availability, production Postgres/Redis deployment.

Absent: production ABDM/JWKS, vendor HIS integration, longitudinal patient reconstruction, historical similarity/trends, raw scan archive/viewer, connected physician OCR queue, caregiver mode, per-fact accept/reject.

7. Feature matrix

Capability	Status	Evidence / current behavior	Gap / action
Longitudinal Patient model	MISSING	No Patient table or patient APIs	Add durable patient and identifier models
Encounter model	PARTIAL	IntakeSession represents one encounter	Promote confirmed session to durable Encounter
ClinicalFact model	PARTIAL	SessionFact is structured but purged	Add durable fact/event rows linked to Encounter
Source provenance	PARTIAL	Rich session spans; FHIR provenance incomplete	Persist evidence; cover every exported clinical resource
Adaptive state machine	DONE	YAML ask_if and deterministic machine	Preserve
SOCRATES branching	DONE	HPI ontology and conditional machine	Preserve; clinician-review content
Voice ASR	PARTIAL	Browser Web Speech; optional adapters	Remove invented confidence; validate Hindi/Tamil and offline claims
Voice TTS	PARTIAL	Browser speech synthesis/server fallback	Validate target devices/languages
Multilingual flow	PARTIAL	10 language choices; English/Hindi content, visible fallback	Complete and validate Tamil demo content
Touch/text fallback	DONE	Independent answer route	Preserve
Low-confidence speech fallback	UNSAFE	Backend threshold exists; browser maps confidence 0 to 0.7	Never synthesize confidence; require clarification
Consent	UNSAFE	Scope gates exist; ownership/ABAC missing on many session routes	Enforce subject/role authorization on every route
Identity/ABHA mock	PARTIAL	Clearly mocked JWT/OTP	Add ownership checks; keep mock label
Document upload	DONE	Session upload endpoint/UI	Add durable encrypted document reference policy
OCR	DONE	Text-layer + Tesseract	Preserve and benchmark
Medication extraction	DONE	OCR entity patterns to facts	Promote to durable MedicationEvent

Capability	Status	Evidence / current behavior	Gap / action
Lab extraction	PARTIAL	Entities extracted; projection drops some units/ranges/dates	Preserve full observation semantics
Bounding-box provenance	PARTIAL	Bbox stored; raw page is not stored/rendered	Store authorized page asset or verifiable reference
Timeline	PARTIAL	Current session documents only	Build cross-encounter event timeline
Medication history	MISSING	No cross-visit query/state	Add dated status/source event view
Symptom recurrence	MISSING	No episode aggregation	Add same-patient recurrence projection
Historical similarity retrieval	MISSING	No embeddings/query; pgvector image unused	Implement explainable same-patient retrieval
Contradiction detection	DONE	Current ledger rules retain both facts	Extend across encounters
Red-flag engine	PARTIAL	Rules run after each answer; proposals not persisted	Persist evaluated/fired decisions
Patient review	DONE	Review and superseding correction	Preserve
Physician dashboard	PARTIAL	Strong current-encounter cockpit	Add longitudinal memory/history panels
Click-to-source	PARTIAL	Transcript source works; document viewer is blank bbox outline	Render original authorized evidence
Physician edit/confirm	PARTIAL	Edit and whole encounter commit	Add fact accept/reject and rejection reasons
Draft summary	DONE	Deterministic and traceability-gated	Keep as one view of longitudinal record
FHIR bundle	PARTIAL	Preview/commit exists; incomplete resource/provenance coverage	Add Allergy/Procedure as valid; provenance per resource
HIS stub	DONE	Local Bundle receiver	Keep explicitly labeled stub
Audit trail	PARTIAL	Hash chain works; red-flag proposals not stored	Close event coverage and test it
Session teardown	DONE	Submit/TTL/revoke purge	Promote confirmed facts before purge transactionally
AYUSH mode	PARTIAL	Opt-in self-report section	Make consultation type visible; longitudinalize
Dashavidha capture	PARTIAL	YAML capture exists; coding helper unused	Wire validated coding and timeline events
Terminology validation	DONE	Closed vocabulary and unmapped	Expand official datasets
Demo mode	PARTIAL	Five real-pipeline cases	Add true multi-visit recurrence case
Evaluation harness	PARTIAL	50 dev + 12 held-out, strict invariants	Add longitudinal, auth, similarity, OCR-UI tests
Offline fallback	PARTIAL	Touch/text and offline extractor work	Browser voice offline status unproven; pin offline eval
Tests	PARTIAL	200 pytest pass; lint/typecheck pass	Missing auth ownership, UI verification, longitudinal tests

8. Reuse plan

Keep the React kiosk, ontology, deterministic machine, `record_fact()` provenance checks, red-flag and contradiction rule engines, OCR backends/entity parser, summary traceability, consent records, audit chain, terminology guard, FHIR builder, demo harness, and most physician components.

Replace or migrate the storage boundary: current `IntakeSession` remains temporary capture, while physician confirmation writes durable `Patient/Encounter/Fact/Evidence/Event` rows before purge. Refactor current projections to query durable encounter data and extend rather than discard the UI.

9. Risks

1. Authorization: many session/dialogue/document routes lack ABAC and patient ownership checks; AUTH_REQUIRED=false is the default.
2. False voice confidence: browser confidence 0 is converted to 0.7, defeating the low-confidence safeguard.
3. Evidence gap: document bytes/pages are not stored, so click-to-source cannot show the original scan.
4. Disconnected OCR review: physician UI initializes pending entities to an empty list and normal responses do not populate it.
5. Documentation overclaim: docs/CURRENT_STATE.md measures an older brief and calls several incomplete integrations DONE.
6. Non-hermetic verification: make check selected Groq because a key was present and stalled on network; LLM_BACKEND=offline completed strict evaluation.
7. FHIR provenance/resource gaps and unused AYUSH coding helper.
8. Clinical validation is absent; all cases are synthetic and red-flag/ontology content needs clinician review.

10. Proposed target architecture

Keep three surfaces. Add a durable clinical domain with Patient, PatientIdentifier, Encounter, ClinicalFact, SourceEvidence, DocumentRecord, MedicationEvent, ObservationEvent, TimelineEvent, Contradiction, RedFlagEvent, PhysicianDecision, FHIRExport, and AuditEvent. Keep temporary CaptureSession separate.

Confirmation transaction: validate draft -> write durable encounter/events/evidence -> build immutable FHIR export -> record physician decision/audit -> optionally push -> purge raw/session artifacts. Patient timeline and similarity services must read only durable confirmed data.

11. Migration plan

1. Fix authorization and false confidence before adding new data surfaces.
2. Add Patient/Encounter/Fact/Evidence migrations and repository APIs.
3. Implement atomic promotion from confirmed session to durable encounter before purge.
4. Add synthetic multi-visit seed patient and patient history endpoints.
5. Rebuild physician timeline, medication and document evidence views over durable data.
6. Add recurrence and deterministic shared-feature retrieval; embeddings only if justified.
7. Complete OCR verification plumbing and original document evidence handling.
8. Close FHIR/AYUSH provenance and coding gaps.
9. Extend evaluation with authorization, cross-visit, purge/promotion, similarity and browser tests.

12. First vertical slice

One synthetic patient with two confirmed historical encounters and one new intake. The new encounter records source-linked facts; the doctor sees prior events, medications, one explainable similar encounter, and original evidence; confirmation writes a durable third encounter, exposes a FHIR preview, and then purges temporary capture. No new LLM is required.

13. Files intended for the next implementation phase

- `app/db/models.py` and a new Alembic migration
- new `app/modules/longitudinal/repository/projection` services
- new patient/encounter API routes mounted in `app/main.py`
- `app/api/routes_physician.py`, `app/modules/consent/session.py`, `app/fhir/bundle.py`
- `frontend/src/physician/PhysicianApp.tsx`, `TimelineView.tsx`, and new medication/similarity panels
- demo fixtures/scripts and new invariant/API/e2e tests

No implementation change is made in this audit.

14. Questions for the project owner

1. Provide the official SIH26047 problem statement/acceptance criteria, because the handoff ranks it above repository behavior but does not attach it.
2. Define the retention policy for confirmed transcript snippets and original document pages needed for click-to-source after raw session purge.
3. Confirm whether the next milestone is the time-short one-language vertical slice or the full English/Hindi/Tamil demo.
4. Nominate a clinician to approve the ontology and red-flag rules before any claim of clinical validation.

Verification performed

- `ruff`, `mypy`, and `TypeScript` checks passed.
- `pytest`: 200 passed.
- Strict offline evaluation: 50/50 development and 12/12 held-out scripts completed; hallucination and unsupported-summary-claim metrics were zero; red-flag sensitivity was 1.0; held-out extraction accuracy was 0.9048.
- `make check` was interrupted because current environment auto-selected Groq and repeated unavailable-network retries. Re-running the strict evaluation with `LLM_BACKEND=offline` passed.
- Repository remained clean before report files were added.

Limitations and disagreements

The official SIH requirements were not provided, so the audit can prove conformity only to the handoff. Browser UI was not launched during this audit; UI conclusions are code-path evidence plus existing automated tests, not a fresh visual run. Percentages use equal weighting and are prioritization aids, not contractual acceptance scores. The handoff leaves retention policy, FHIR profiles, similarity thresholds, auth roles, language quality, and clinical validation thresholds unspecified.

Claim-to-source ledger

- Handoff product/longitudinal requirements: `SIH26047_CODEX_HANDOFF_FULL_CONTEXT_AND_AUDIT.md`, project owner, 24 Aug 2026, lines 93-157, 281-322, 379-428, 564-705, 1252-1408, 1562-1583.
- Handoff audit checklist/format: same source, lines 710-1082 and 1641-1697.
- Session/durable model and purge semantics: `app/db/models.py`, lines 1-11 and 129-219; `app/modules/consent/session.py`, lines 154-245.

- Current adaptive/extraction/summary behavior: `app/modules/dialogue/machine.py`, lines 168-405; `app/llm/extraction.py`, lines 139-209; `app/modules/summary/generate.py`, lines 57-87.
- Current document pipeline and timeline: `app/api/routes_documents.py`, lines 31-150; `app/modules/documents/pipeline.py`, lines 153-260.
- Voice confidence issue: `frontend/src/shared/useSpeech.ts`, lines 163-167; `app/speech/client.py`, lines 28-39.
- Authorization gaps: unguarded routes in `app/api/routes_session.py`, `routes_dialogue.py`, and `routes_documents.py`; anonymous default in `app/auth/identity.py`, lines 58-62.
- FHIR coverage: `app/fhir/bundle.py`, lines 154-307; test coverage in `tests/test_api_end_to_end.py`, lines 291-297.
- Verification results: local command output on 24 Aug 2026; `docs/EVALUATION.md` for metric definitions.